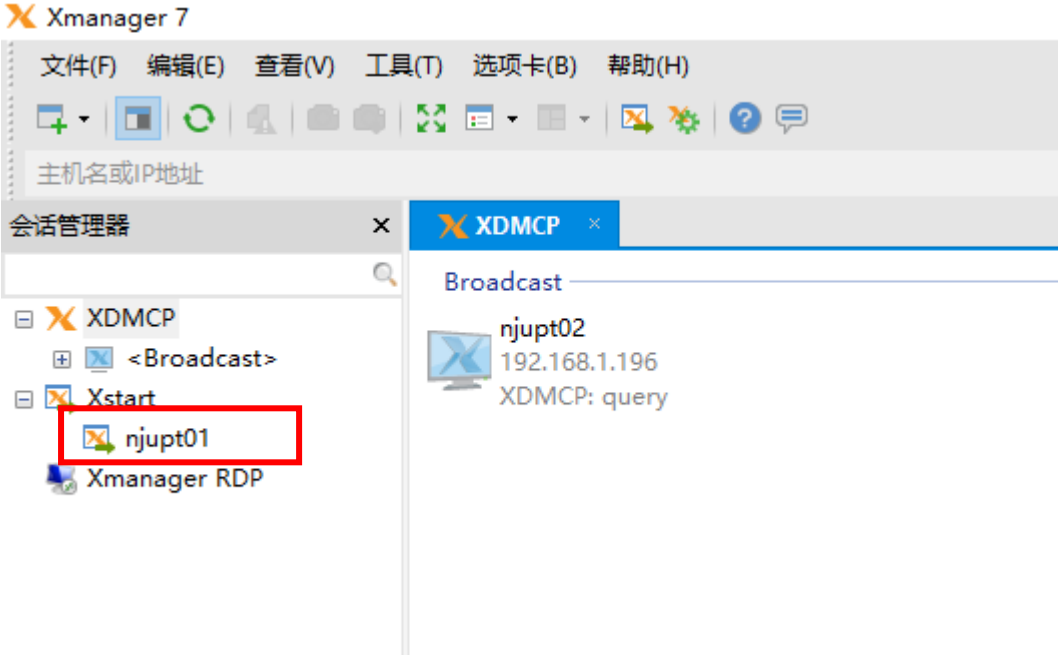
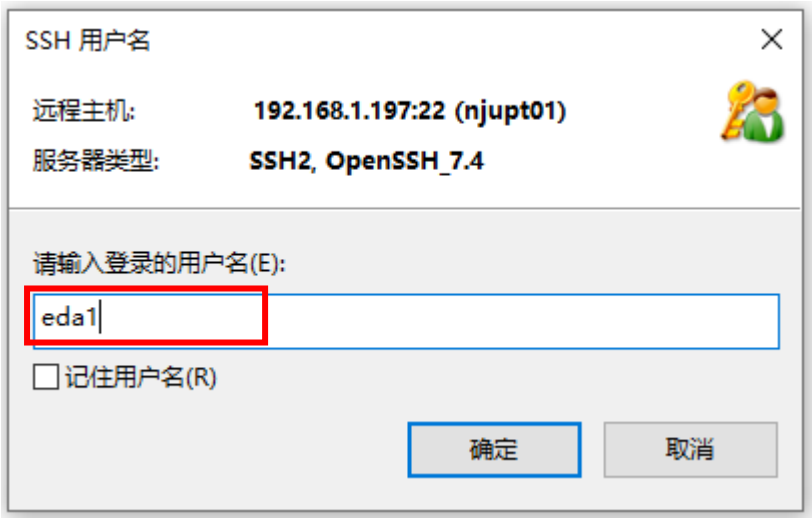
课程设计软件使用说明

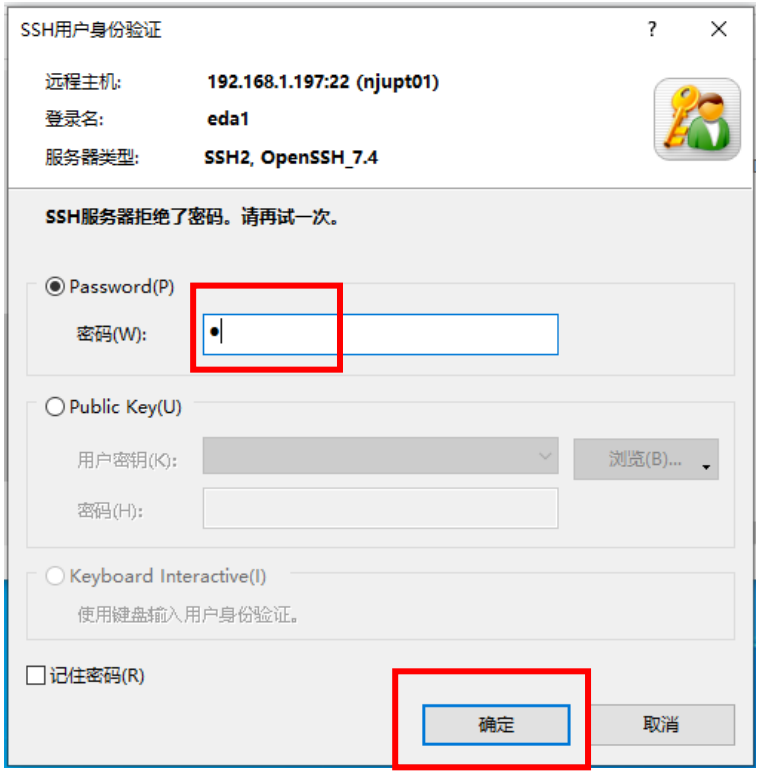
1. 打开桌面 Xmanager

2. 双击“会话管理器”— Xstart –- njupt01

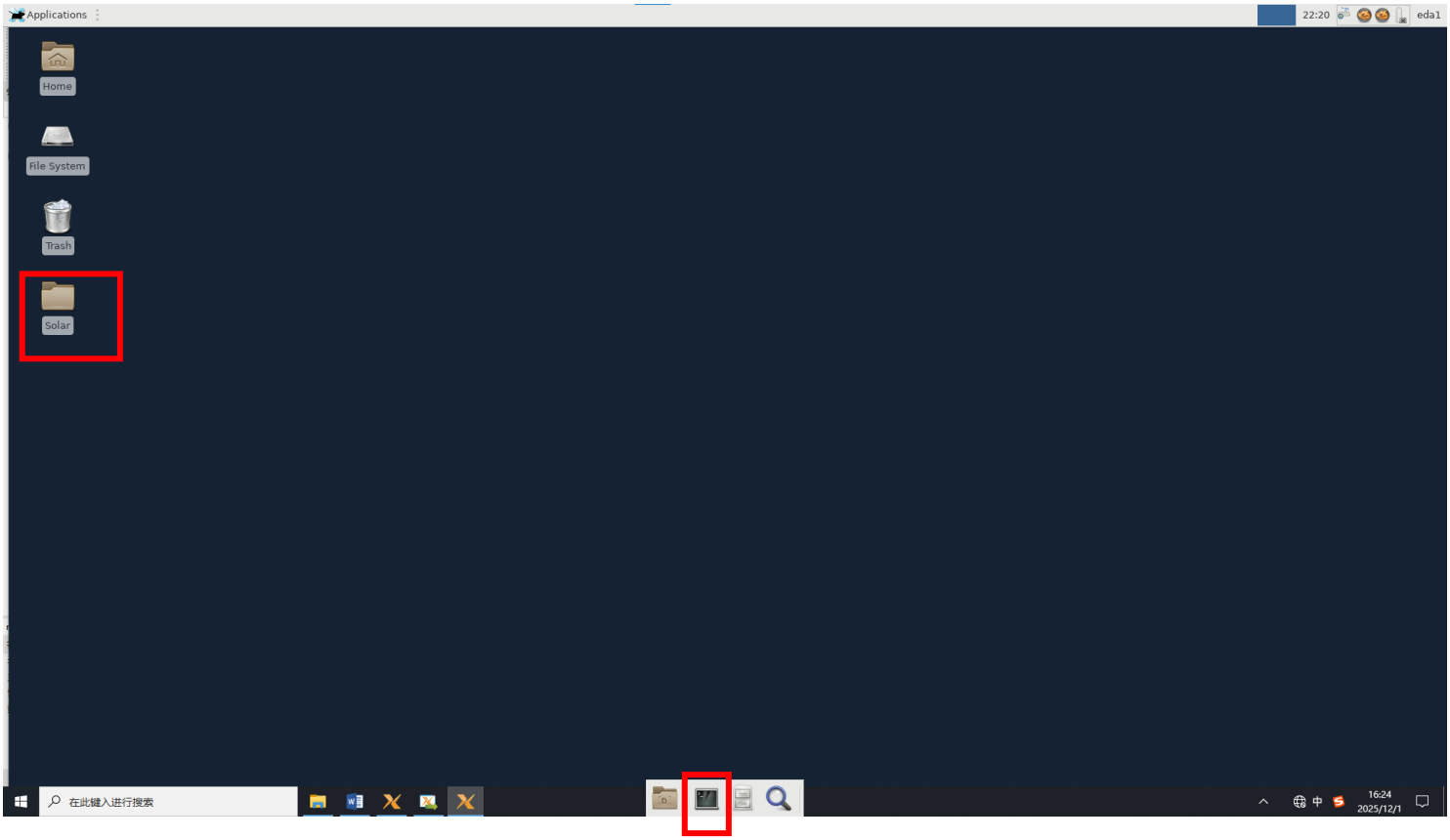


3. 输入用户名“eda 学号”，如学号为 1，则用户名为“eda1”，密码 1，确定。

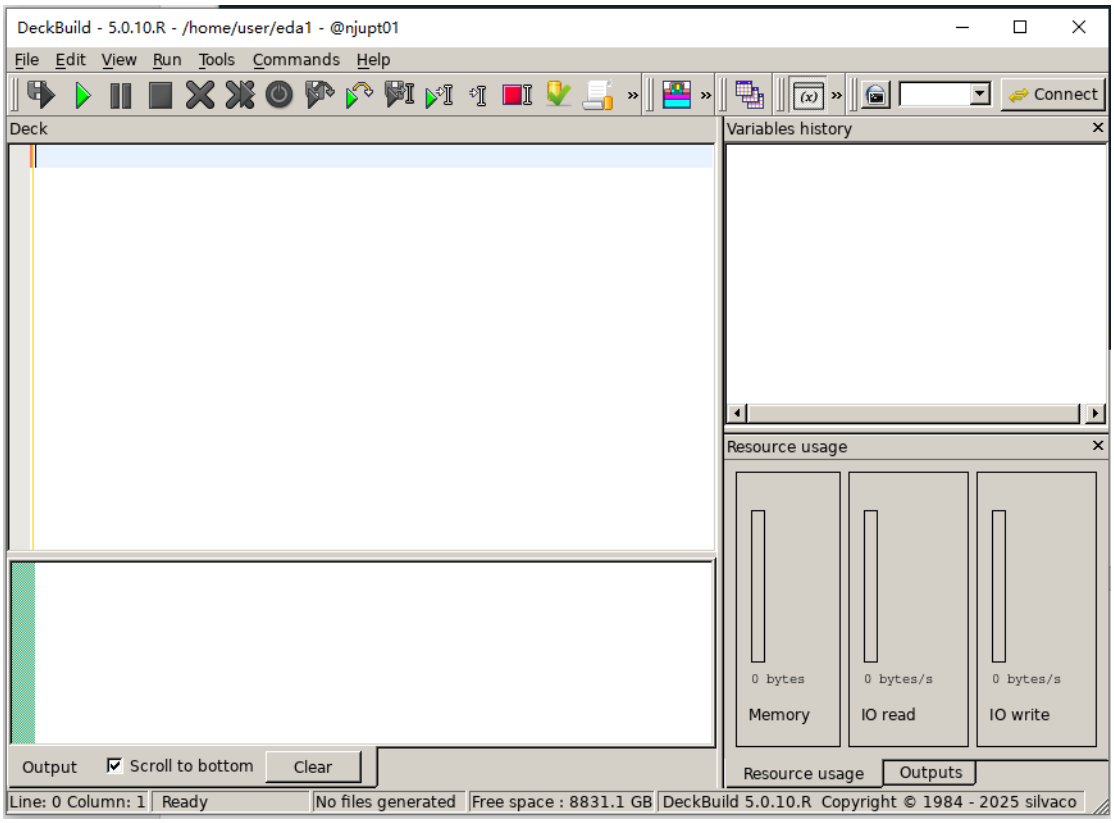




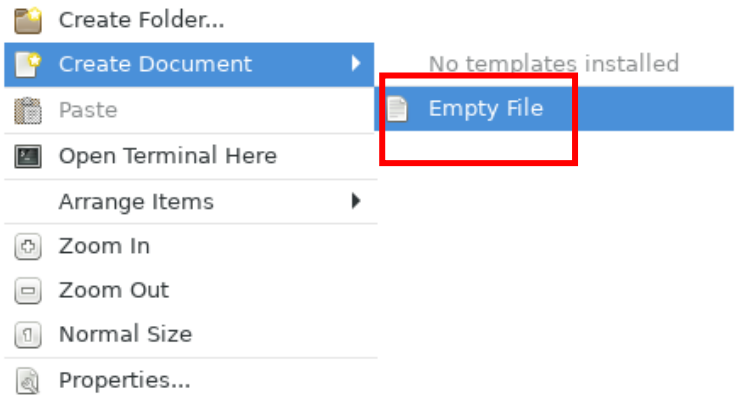
4. 可在桌面新建项目文件夹（如 Solar），并打开屏幕下方终端 Terminal。



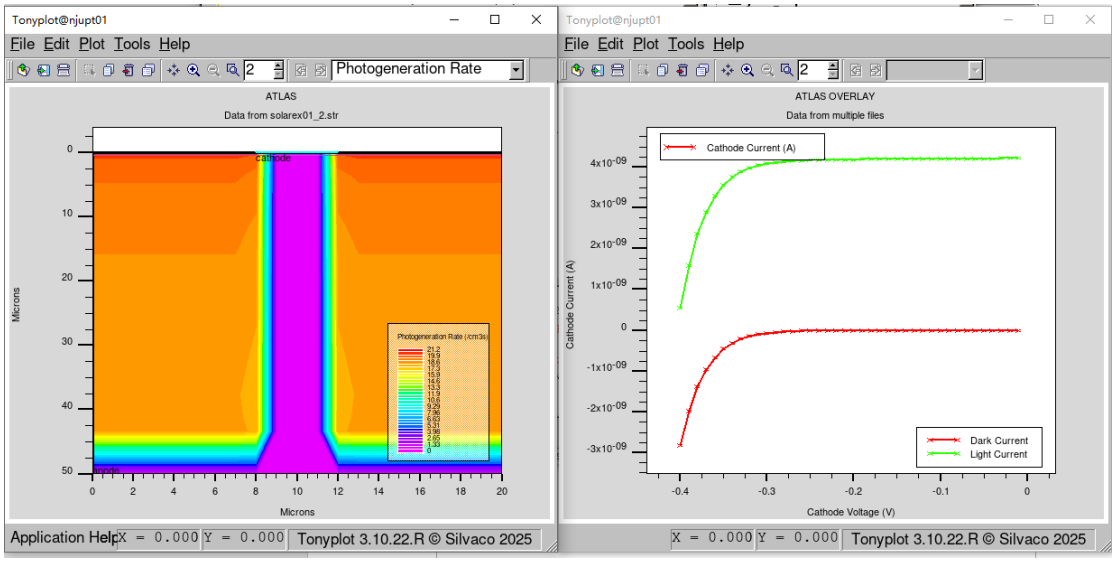
5. 终端输入 source /home/eda/bashrc，再输入 deckbuild 启动 Silvaco 仿真器：

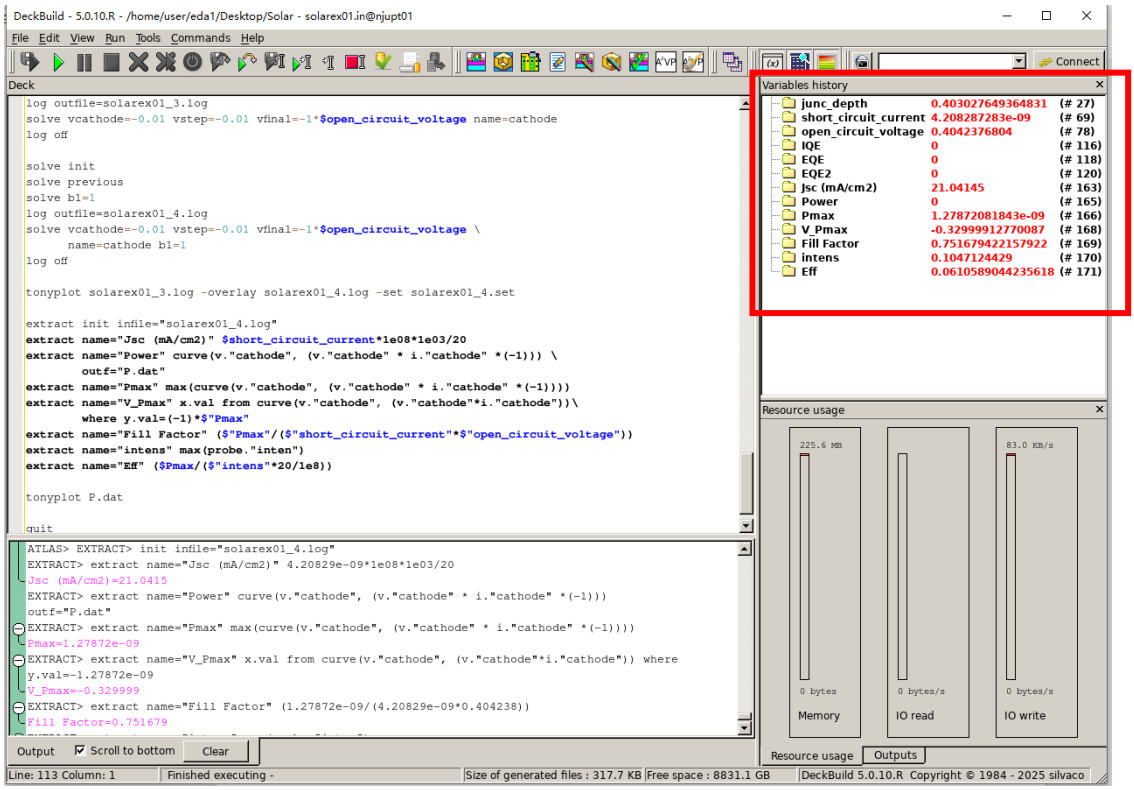


6. 在项目文件夹中复制所给代码文件（可右键新建空文件，复制代码后保存为相应格式即可）。



7. 使用 DeckBuild 打开.in 文件，并修改如下一条语句防止报错：
原始：extract init inf="solarex01_2.log"
修改后：extract init infile="solarex01_2.log"


Tonyplot@njupt01
File Edit Plot Tools Help
Photogeneration Rate
ATLAS
Data from solarex01_2.str
Cathode
Microns
Application Help
X = 0.000 Y = 0.000
Tonyplot 3.10.22.R © Silvaco 2025
ATLAS OVERLAY
Data from multiple files
Cathode Current (A)
Dark Current
Light Current
Cathode Voltage (V)


DeckBuild - 5.0.10.R - /home/user/eda1/Desktop/Solar - solarex01.in@njupt01
File Edit View Run Tools Commands Help
Deck
log outfile=solarex01_3.log
solve vcathode=-0.01 vstep=-0.01 vfinal=-1*$open_circuit_voltage name=cathode
log off
solve init
solve previous
solve b1=1
log outfile=solarex01_4.log
solve vcathode=-0.01 vstep=-0.01 vfinal=-1*$open_circuit_voltage \
name=cathode b1=1
log off
tonyplot solarex01_3.log -overlay solarex01_4.log -set solarex01_4.set
extract init infile="solarex01_4.log"
extract name="Jsc (mA/cm2)" $short_circuit_current*1e08*1e03/20
extract name="Power" curve(v."cathode", (v."cathode" * i."cathode" *(-1))) \
outf="P.dat"
extract name="Pmax" max(curve(v."cathode", (v."cathode" * i."cathode" *(-1))))
extract name="V_Pmax" x.val from curve(v."cathode", (v."cathode"*i."cathode"))\
where y.val=(-1)*$"Pmax"
extract name="Fill Factor" ($"Pmax"/($"short_circuit_current"*$"open_circuit_voltage"))
extract name="intens" max(probe."inten")
extract name="Eff" ($Pmax/($"intens"*20/1e8))
tonyplot P.dat
quit
ATLAS> EXTRACT> init infile="solarex01_4.log"
EXTRACT> extract name="Jsc (mA/cm2)" 4.20829e-09*1e08*1e03/20
Jsc (mA/cm2)=21.0415
EXTRACT> extract name="Power" curve(v."cathode", (v."cathode" * i."cathode" *(-1)))
outf="P.dat"
EXTRACT> extract name="Pmax" max(curve(v."cathode", (v."cathode" * i."cathode" *(-1))))
Pmax=1.27872e-09
EXTRACT> extract name="V_Pmax" x.val from curve(v."cathode", (v."cathode"*i."cathode")) where
y.val=-1.27872e-09
V_Pmax=-0.329999
EXTRACT> extract name="Fill Factor" (1.27872e-09/(4.20829e-09*0.404238))
Fill Factor=0.751679
Output Scroll to bottom Clear
Variables history
junc_depth 0.403027649364831 (# 27)
short_circuit_current 4.208287283e-09 (# 69)
open_circuit_voltage 0.4042376804 (# 78)
IQE 0 (# 116)
EQE 0 (# 118)
EQE2 0 (# 120)
Jsc (mA/cm2) 21.04145 (# 163)
Power 0 (# 165)
Pmax 1.27872081843e-09 (# 166)
V_Pmax -0.32999912770087 (# 168)
Fill Factor 0.751679422157922 (# 169)
intens 0.1047124429 (# 170)
Eff 0.0610589044235618 (# 171)
Resource usage
Memory
IO read
IO write
Resource usage Outputs
Line: 113 Column: 1
Finished executing -
Size of generated files : 317.7 KB
Free space : 8831.1 GB
DeckBuild 5.0.10.R Copyright © 1984 - 2025 silvaco